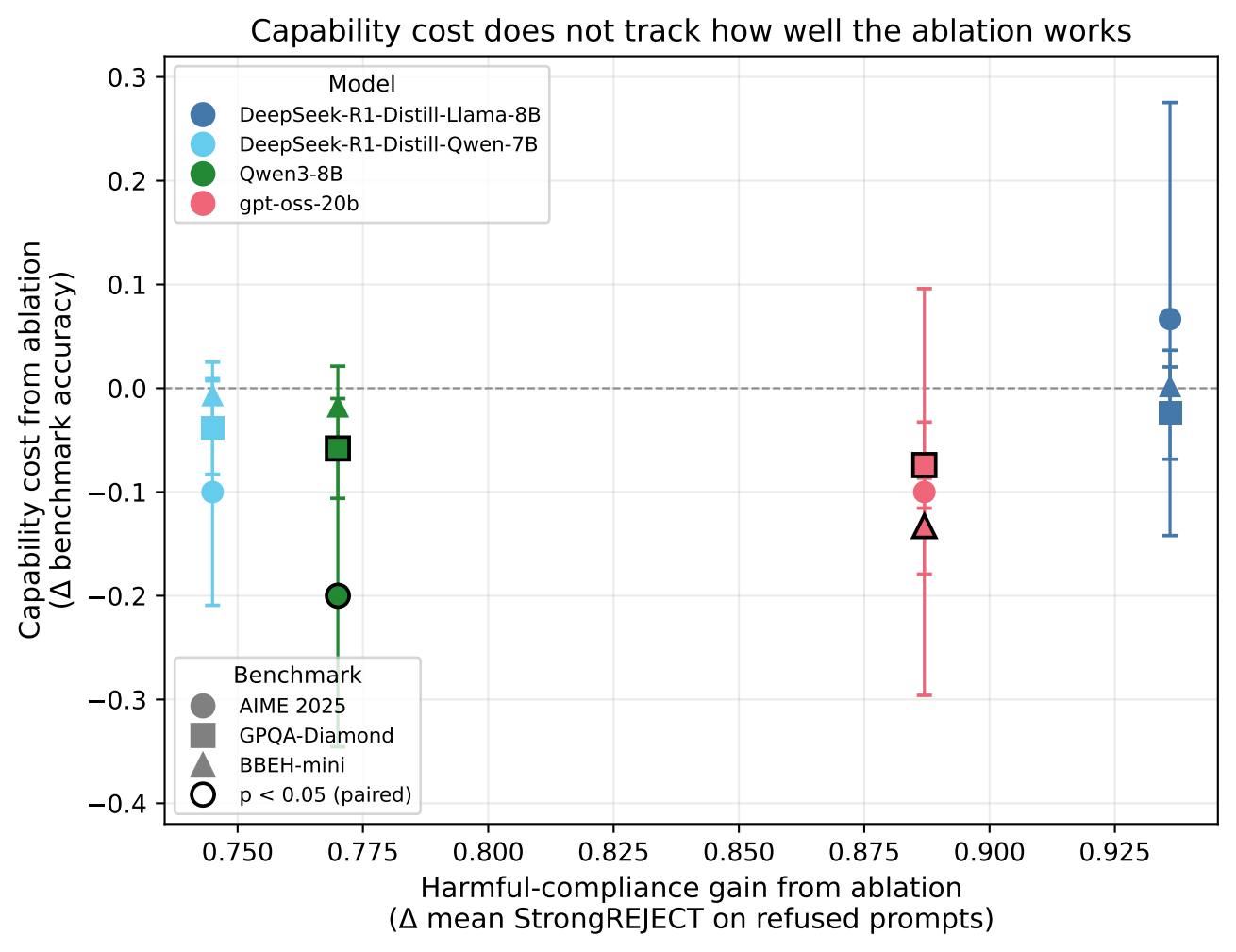
Capability cost does not track how well the ablation works
Model
DeepSeek-R1-Distill-Llama-8B
DeepSeek-R1-Distill-Qwen-7B
Qwen3-8B
gpt-oss-20b
Benchmark
AIME 2025
GPQA-Diamond
BBEH-mini
p < 0.05 (paired)
Capability cost from ablation
(Δ benchmark accuracy)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0.750
0.775
0.800
0.825
0.850
0.875
0.900
0.925
Harmful-compliance gain from ablation
(Δ mean StrongREJECT on refused prompts)